

Matthew Kim

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EDUCATION

University of Virginia

Charlottesville, VA

B.A. in Applied Statistics, Minor in Computer Science | GPA: 3.9/4.0

Aug. 2023 – May 2027

- **Data & Analytics:** Monte Carlo simulation, bootstrap methods, hypothesis testing, EDA, ANOVA, regression, cross-validation, pandas, NumPy, Jupyter, data visualization
- **Computer Science:** Python, Java, C, JavaScript, TypeScript, React, Next.js, Flask, Tailwind CSS, SQLite, Git/GitHub, REST APIs

EXPERIENCE

Research Assistant

2025 – Present

UVA Historic Fashion Archive

Charlottesville, VA

- Architecting a Next.js 16 / React 19 / TypeScript frontend for UVA's vintage clothing collection, integrating a CollectiveAccess CMS via a custom API client (cookie + Basic Auth token flow, 5-minute in-memory cache, static JSON fallback) to ensure data availability under varying backend conditions
- Implementing content-based filtering scoring garment similarity across five weighted attributes to power recommendations; built paginated collection pipeline with multi-attribute filtering, boolean search, and export to JSON/CSV/PDF

Teaching Assistant

January 2026 – Present

University of Virginia — CS1112

Charlottesville, VA

- Supported 30+ students in introductory Python through weekly office hours, lab sections, and one-on-one debugging; graded assignments and exams with structured feedback on data structures, control flow, and functions

PROJECTS

Hana-Bi: E-Commerce Platform | *Next.js, TypeScript, Stripe, Resend, Zustand*

2024 – Present

- Engineering a full-stack e-commerce platform (Next.js 15, TypeScript, Stripe, Resend) with Stripe as the product data source of truth — implementing an autopaging catalog pipeline, per-product metadata schema, and serverless-safe session verification via direct Stripe API calls
- Building end-to-end transactional data flow: client cart state (Zustand), Checkout Session creation, webhook-driven order confirmation emails via Resend, and a waitlist capture pipeline with dual-notification routing

Tech Industry Layoff Analysis | *R, tidyverse, ggplot2, R Markdown*

2026

- Analyzed 2,700+ tech layoffs using R and tidyverse, applying t-tests and ANOVA to compare workforce reduction severity across company types; built visualizations in ggplot2 uncovering that private companies cut proportionally twice as much of their workforce as public ones (32.4% vs. 16.7%)
- Collaborated in a 3-person team using GitHub for version control, delivering a fully reproducible R Markdown report

Insurance Client Data Management System | *Python, Flask, Google Sheets API, Render*

2025

- Built a Flask web app with a minimal intake form syncing entries to Google Sheets via the Sheets API (deployed on Render); replaced a binder-based workflow covering ~2,000 records with searchable, sortable data and CSV export

NBA Salary Prediction Model | *R, multiple linear regression, cross-validation*

2025

- Built a multiple linear regression model predicting NBA player salaries from performance, awards, age, and position; achieved $\text{adj. } R^2 \approx 0.77$ via five-fold cross-validation; identified Points, Minutes, Rebounds, and Assists as dominant predictors via Cook's distance and residual analysis

TECHNICAL SKILLS

Languages: Python, R, Java, C, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Next.js, Flask, Tailwind CSS, pandas, NumPy, tidyverse, ggplot2, Three.js

Developer Tools: Git/GitHub, Stripe API, Google Sheets API, Resend, Zustand, Render, Jupyter, REST APIs

Statistical Methods: Hypothesis testing, ANOVA, regression, bootstrap, Monte Carlo simulation, cross-validation, EDA